Prouver l'exécution d'un automate (STARK)

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Sommaire

Sommaire

Problématique

Question scientifique

Comment prouver la connaissance d'un secret, de façon vérifiable et efficace, sans le révéler ?

- Cas d'étude: connaissance d'un mot reconnu par un automate
- Enjeu: preuve courte, vérification rapide

Objectif

- Automate A connu publiquement
- Prouver: connaître un mot w reconnu par A
- Sans divulguer w

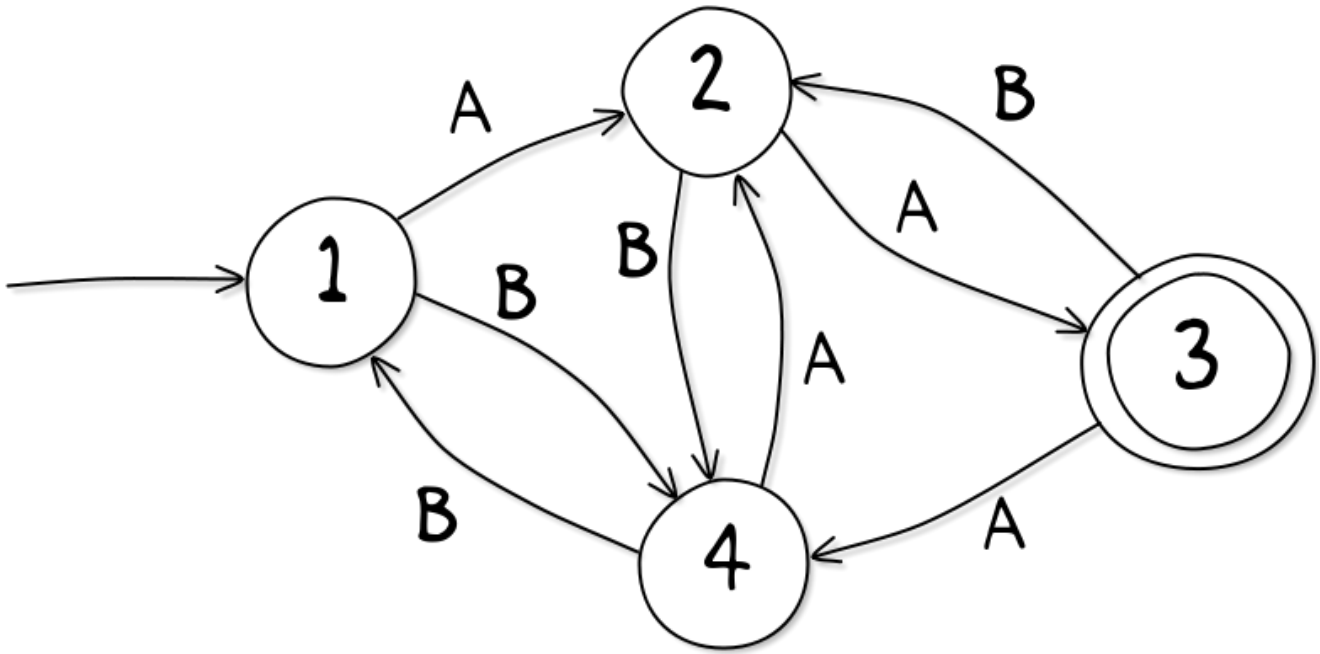


Figure: Un exemple d'automate (csfieldguide.org.nz)

Motivations

- Prouver son identité sans rien révéler de plus

Sommaire

Prérequis: Fonction de hachage

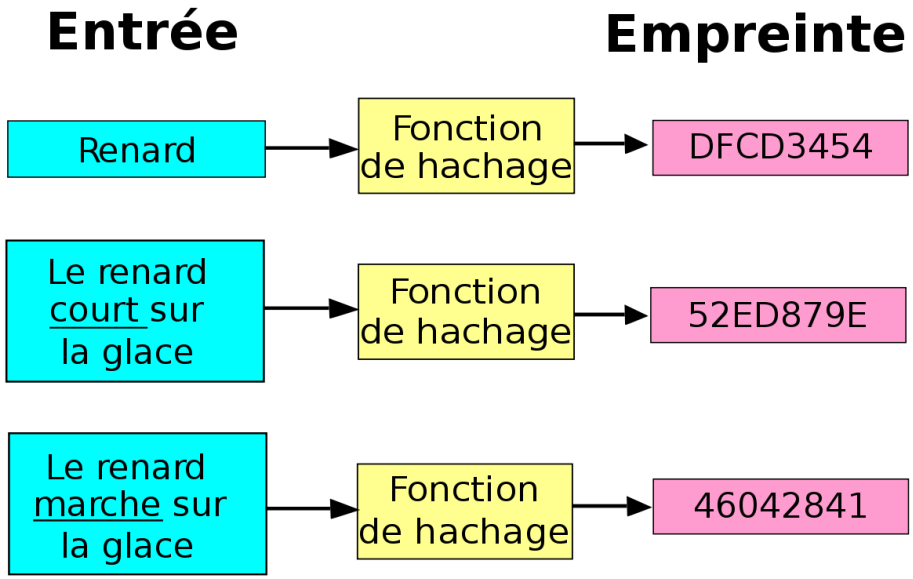
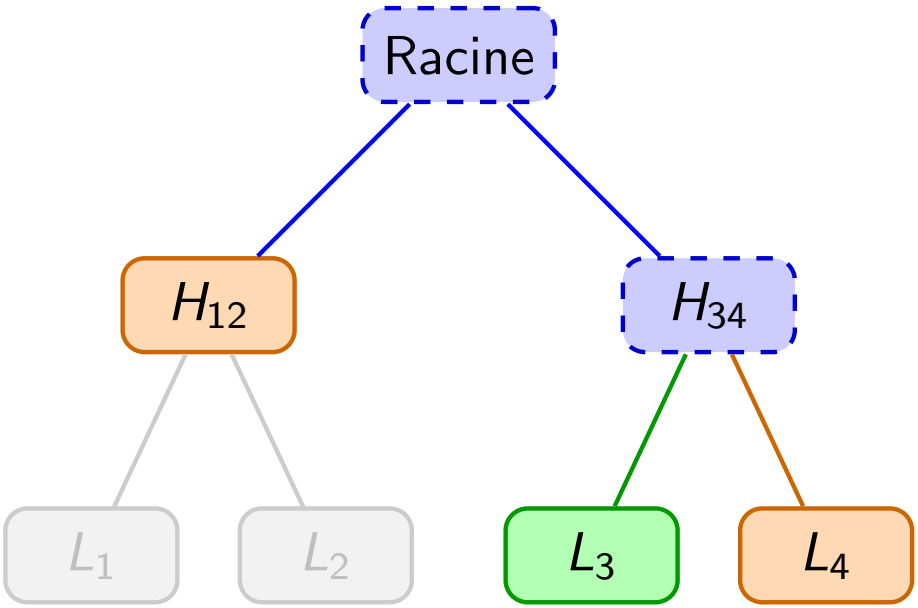


Figure: Illustration d'une fonction de hachage (techno-science.net)

Critères essentiels:

1. Résistance aux collisions
2. Résistance à la préimage
3. Résistance à la deuxième préimage
4. Distribution uniforme

Prérequis: Arbres de Merkle



- Preuve d'appartenance: **nœuds frères** seulement
- Recalcul itératif (**bleu**) jusqu'à la racine
- Taille de preuve: $\mathcal{O}(\log_2 N)$

Reconstruction

$$H_{\text{parent}} = \text{Hash}(H_g \parallel H_d)$$

Prérequis: L'heuristique de Fiat-Shamir

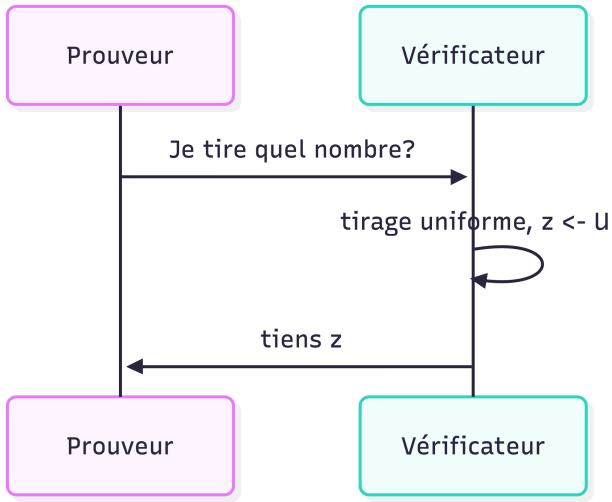


Figure: Version intéractive

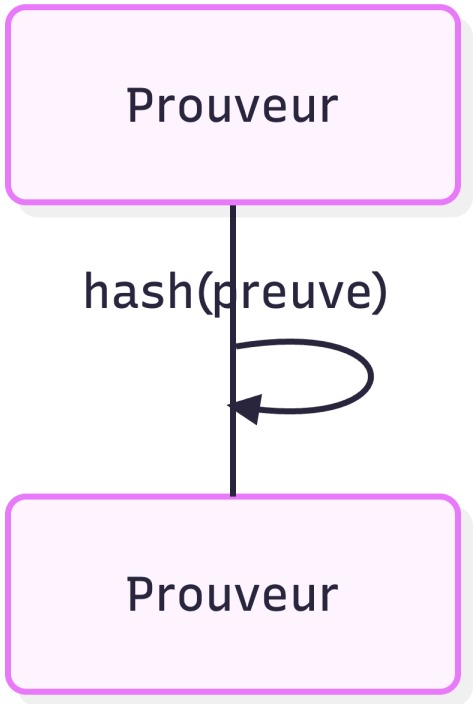


Figure: Version non intéractive

Sommaire

Architecture ZK-STARK

- **Prouveur**: forte puissance de calcul, données privées
- **Vérificateur**: accès public, faible puissance de calcul
- Cas d'usage: vérification de programmes sur une blockchain publique

Prouver l'exécution d'un automate

Déclaration naïve

Je connais un mot W accepté par un automate A

- Majuscules: variables publiques
- Minuscules: connues du seul prouveur
- Idée: prouver que chaque transition est correcte

Sommaire

AIR - Algebraic Intermediate Representation

- AIR: transformer un calcul en problème algébrique
- But: prouver la validité de chaque transition

étape	état	caractère
0	1	B
1	4	B
2	1	A
3	2	B
4	4	A
5	2	ε

Table: Table de transitions

AIR - Algebraic Intermediate Representation

- Un polynôme par colonne (P_i, P_e, P_c), via interpolation de Lagrange

$$P_e = \frac{11x^5}{120} - \frac{35x^4}{24} + \frac{65x^3}{8} - \frac{445x^2}{24} + \frac{887x}{60} + 1$$

3 contraintes à respecter:

1. L'exécution démarre à l'état initial
2. Chaque transition est valide (est dans la définition de l'automate donné en paramètre)
3. L'exécution termine sur un état final

AIR - Algebraic Intermediate Representation

Contraintes $\rightarrow$ polynômes, via:

$$P(x) = 0 \iff (X - x) | P \iff \frac{P}{X - x} \text{ est un polynôme}$$

1. $C_0 = \frac{P_e(0) - q_0}{X}$
2. $C_1 = \frac{P_e(X+1) - P_t(P_e(X), P_c(X))}{\prod_{0 \leq i \leq n-1} (X - i)}$
3. $C_2 = \frac{\prod_{q \in F} P_e(n-1) - q}{X - (n-1)}$

P_t: polynôme de transition (état, caractère) $\rightarrow$ état suivant

CP - Polynôme de composition

Combiner les contraintes en un seul polynôme, pour appliquer FRI:

$$CP = \alpha_0 C_0 + \alpha_1 C_1 + \alpha_2 C_2$$

- $\alpha_0, \alpha_1, \alpha_2$: tirés aléatoirement par le vérificateur
- Garantissent $CP = 0 \iff C_0 = C_1 = C_2 = 0$ (en probabilités), même si le vérificateur choisit les coefficients

Sommaire

Protocole FRI: Fast Reed-Solomon Interactive Oracle Proof of Proximity

- But: montrer que CP s'annule sur toutes les étapes de calcul
- Solution naïve: envoyer le polynôme et tous les points
- Problème: aucune compression, aucune confidentialité

Protocole FRI

Objectif

Montrer que CP est un polynôme, en le ramenant degré par degré ($D \to D/2 \to \dots$) jusqu'à un polynôme constant.

$$\Delta(f, P) = \frac{|x \in D | f(x) \neq P(x)|}{|D|}$$

- On montre que f est δ-proche d'un polynôme: $P(\Delta(f, P) < \delta) \approx 1$
- L'opérateur FRI divise le degré par 2 sans perdre cette propriété
- Le vérificateur n'a plus qu'à vérifier des points alignés

Comment fonctionne FRI

Inspiration

Même idée que la FFT pour les polynômes.

1. On sépare le polynôme en un polynôme avec les puissances paires, et un autre avec les impaires $P(X) = p(X^2) + Xi(X^2)$
2. On tire un β aléatoirement (Fiat-Shamir)
3. On considère maintenant $\tilde{P} = p(X) + \beta i(X)$

Commitment

Il faut attester chacune des applications de l'opérateur FRI.

Sommaire

État des lieux

Voici ce que connaît le vérificateur:

1. l'engagement de CP
2. l'engagement de chaque application de FRI
3. tous les tirages aléatoires

Preuve au complet

1. Générer la trace de l'exécution de l'automate et de la fonction de hachage
2. Créer tous les polynômes de représentation intermédiaires
3. Évaluer ces polynômes et les engager
4. Créer CP (donc vérifier les contraintes)
5. Appliquer FRI sur CP itérativement
6. Répeter la dernière étape autant de fois qu'on veut (nombre de points d'évaluation)
7. Envoyer des métadonnées, les engagements, preuves d'engagements au vérificateur

Vérification

Idée

L'idée est de refaire le processus inverse qu'à fait le prouver mais au lieu de calculer les polynômes on vérifie que les évaluations correspondent.

1. Récupérer les métadonnées, engagement et preuves d'engagements
2. Pour chaque point d'évaluation
   1. Vérifier l'engagement pour l'état, le caractère, et la transition
   2. Vérifier les engagements du hash
   3. Reconstruire **l'évaluation** de CP
   4. Pour chaque application de FRI
      1. Vérifier les engagements
      2. Appliquer FRI au point
      3. Vérifier que le résultat est correct

Prouver l'exécution d'un automate

Déclaration correcte

Je connais un mot w accepté par un automate A tel que $h(w) = H$

- A, H: publics — w: secret du prouveur

Poséidon: une fonction de hachage "ZK-friendly"

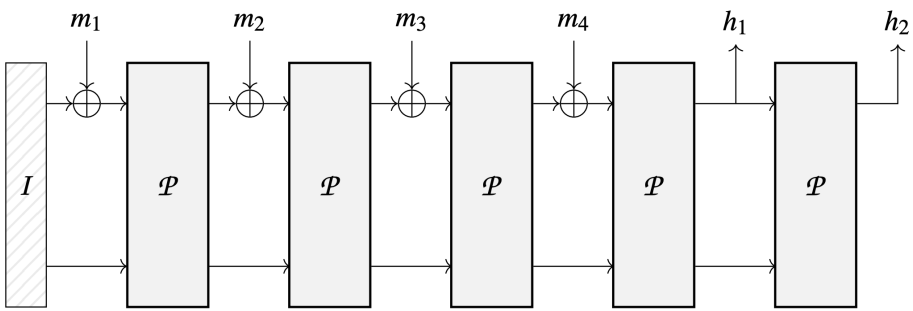


Figure: Une fonction de hachage en éponge

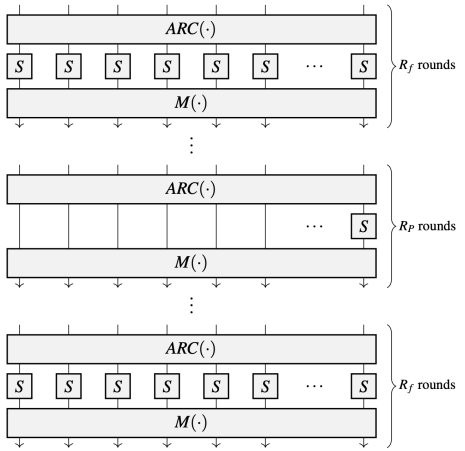


Figure: La permutation poséidon

Merci de votre attention

Sommaire

Annexe: Avancement

1. 01-02/25: Lecture des papiers sur Cairo, STARK et FRI
2. 03/25: STARK 101
3. 04-06/25: Polynômes et utilitaires en Ocaml
4. 09/25: Idée AFD et premier prototype d'AIR
5. 11/25: Prototype fini sans ZK
6. 01/26: Fini la rédaction d'articles sur ce premier prototype
7. 02/26: Fini prototype avec le hash
8. 02/26: Gros problème

Annexe: Code

```
CONSTRAINTS_LENGTH=29
N_QUERY=12
T=8
r_f = 8
r_p = 57
blowup_factor = 128 # WARNING: needs to be a power of two
```

Annexe: Code

```
136
137 class Sponge():
138     def __init__(self, rate, capacity, permutation, r_f, r_p, rc):
139         self.r = rate
140         self.c = capacity
141         self.pi = permutation
142         self.state = None
143         self.r_f = r_f
144         self.r_p = r_p
145         self.mds = generate_cauchy_matrix(rate+capacity)
146         self.rc = ARC
147         self.trace = []
148
149     def apply(self,data):
150         self.state = np.array([FieldElement(0) for i in range(0,self.c+self.r)])
151         self.trace = [self.state]
152         #remaining = self.pad(remaining)
153         chunks = [data[self.r*i:self.r*(i+1)] for i in range(0,len(data)//self.r)]
154         i = 0
155         while(len(chunks) > 0):
156             i+=1
157             self.state, trace = self.pi(self.state + np.pad(chunks[0], (0,self.c), mode='constant'), self.r_f, self.r_p, self.mds, self.rc)
158             chunks = chunks[1:]
159             self.trace.extend(trace)
160         return self.state[0:self.c], self.trace
161
162
163 def generate_cauchy_matrix(t):
164     x = [FieldElement(e) for e in range(0,t)]
165     y = [FieldElement(e) for e in range(t, 2*t)]
166     return np.array([[FieldElement(1)/(x[i]-y[j]) for j in range(0,len(y))] for i in range(0,len(x))])
167
168 def arc(state, rc):
169     return np.array(state) + rc
170
171 def mix(state, mds):
172     return mds @ np.array(state)
173
174 def full_s_box(state):
175     return np.array(state) ** 5
176
177 def partial_s_box(state):
178     return [state[0] ** 5, *state[1:]]
179
```

Annexe: Code

```
180 def hades_round_permutation(state, r_f, r_p, mds, rc):
181     trace = []
182     for r in range(0,r_f+r_p):
183         state = arc(state, rc[r])
184         if(r_f/2 <= r < r_f/2+r_p):
185             state = partial_s_box(state)
186         else:
187             state = full_s_box(state)
188         state = mix(state, mds)
189         trace.append(state)
190     return state, trace
191
192 class Poseidon():
193     def __init__(self):
194         self.sponge = Sponge(7,1,hades_round_permutation,8,57, ARC)
195
196     def hash(self, data):
197         return self.sponge.apply(data)
```

Annexe: Code

```python
import json
import math

# Source - https://stackoverflow.com/a/3035188
# Posted by Robert William Hanks, modified by community. See post 'Timeline' for change history
# Retrieved 2026-02-02, License - CC BY-SA 4.0

def primes(n):
    """ Returns  a list of primes < n """
    sieve = [True] * n
    for i in range(3,int(n**0.5)+1,2):
        if sieve[i]:
            sieve[i*i::2*i]=[False]*((n-i*i-1)//(2*i)+1)
    return [2] + [i for i in range(3,n,2) if sieve[i]]

# ================================================

class Automata:
    def __init__(self, states, alphabet, transition, start_state, accept_states):
        self.states = sorted(list(states))
        self.alphabet = sorted(list(alphabet))
        self.transition = transition
        self.start_state = start_state
        self.accept_states = accept_states
        n_needed = len(self.states) + len(self.alphabet)
        safe_bound = int(n_needed * (math.log(n_needed) + 4)) + 20 if n_needed > 2 else 20
        p_list = primes(safe_bound)[:n_needed]

        # 2. Construct numeric maps
        self.state_map = {state: p_list[i] for i, state in enumerate(self.states)}
        self.caracter_map = {char: p_list[len(self.states) + i] for i, char in enumerate(self.alphabet)}

    @classmethod
    def from_json(cls, json_string):
        """Creates an instance of Automata from a JSON string."""
        data = json.loads(json_string)
        return cls(
            states=set(data["states"]),
            alphabet=set(data["alphabet"]),
            transition=data["transition"],
            start_state=data["start_state"],
            accept_states=set(data["accept_states"])
        )

```

Annexe: Code

```python
    def to_json(self):
        """Serializes the automata instance into a JSON string."""
        data = {
            "states": sorted(list(self.states)),
            "alphabet": sorted(list(self.alphabet)),
            "transition": self.transition,
            "start_state": self.start_state,
            "accept_states": sorted(list(self.accept_states))
        }
        return json.dumps(data)

    def generate_trace(self, input_string):
      trace = {'i':[], 's': [], 'c': []}
      current_state = self.start_state
      i = 0
      for symbol in input_string:
          if symbol not in self.alphabet:
              return False
          trace['i'].append(i)
          i+=1
          trace['s'].append(current_state)
          trace['c'].append(symbol)
          current_state = self.transition[current_state][symbol]

      return trace
```

Annexe: Code

```
#word = open("text.txt", "r").read()
word = "I think STARKS are fun to learn"
```

```
import pandas as pd
from itertools import product
import time
from poseidon import Poseidon, ARC, generate_cauchy_matrix
from field import FieldElement
import numpy as np
from polynomial import Polynomial, interpolate_poly
from channel import Channel
from merkle import MerkleTree, verify_decommitment
from functools import reduce
from constants import N_QUERY, r_f, r_p, blowup_factor, T
import constants
from utils import Automata, primes
from tqdm import tqdm
```

File display

```
start_all = time.time()
```

Annexe: Code

```python
import time
import math

def print_prover_success_banner(start_time, channel, POSEIDON_TRACE_EVAL_LENGTH, STATE_EVAL_LENGTH):
    # 1. Calculations
    duration = time.time() - start_time

    # Proof Size
    proof_size_bytes = sum(len(entry.encode('utf-8')) for entry in channel.proof)
    proof_size_kb = proof_size_bytes / 1024

    # Theoretical Soundness (The claim you are making)
    # Max degree is determined by the Poseidon constraint (degree 5)
    max_degree = (POSEIDON_TRACE_EVAL_LENGTH - 1) * 5
    rho = max_degree / STATE_EVAL_LENGTH
    p_cheat = rho ** N_QUERY
    security_bits = -math.log2(p_cheat) if p_cheat > 0 else 0

    # Expansion Factor (Blowup)
    blowup = STATE_EVAL_LENGTH / POSEIDON_TRACE_EVAL_LENGTH

    # 2. Colors & Art
    BLUE = "\033[94m"
    GREEN = "\033[92m"
    YELLOW = "\033[93m"
    MAGENTA = "\033[95m"
    CYAN = "\033[96m"
    RESET = "\033[0m"
    BOLD = "\033[1m"

    ascii_art = f"""
{BLUE}{BOLD}
  ____   ____   ___   ___   ____    ____  ____  _   _
 |  _ \|  _ \ / _ \ / _ \|  _ \  / ___|  _ \| | | |
 | |_) | |_) | | | | | | | | | | \___ \| |_) | |_| |
 |  __/|  _ /| |_| | |_| | |_| |  ___) |  __/|  _  |
 |_|   |_| \_\\___/ \___/|____/  |____/|_|   |_| |_|

{RESET}"""

    print(ascii_art)
    print(f"{BOLD}Status:{RESET}   {BLUE}PROOF GENERATED SUCCESSFULLY{RESET}")
    print("-" * 60)
    print(f"{BOLD}Computational Effort:{RESET}")
    print(f"  ⏱️  {YELLOW}Generation Time:{RESET}  {duration:.4f} seconds")
    print(f"  📈 {CYAN}Trace Length:{RESET}     {POSEIDON_TRACE_EVAL_LENGTH} steps")
    print(f"  🔍 {CYAN}Blowup Factor:{RESET}    {blowup:.1f}x (Domain size: {STATE_EVAL_LENGTH})")
    print("-" * 60)
    print(f"{BOLD}Proof Artifacts:{RESET}")
    print(f"  📦 {GREEN}Proof Size:{RESET}       {proof_size_kb:.2f} KB")
    print(f"  🛡️  {MAGENTA}Claimed Security:{RESET} {security_bits:.1f} bits (P(Cheat) ≈ {p_cheat:.1e})")
    print("-" * 60)
```

File display

Annexe: Code

```
automata = Automata.from_json(open("automata.json", "r").read())
trace = automata.generate_trace(word)
```

```
mds = generate_cauchy_matrix(T)
```

```
hash_input = [FieldElement(x) for x in np.pad([automata.caracter_map[x] for x in word], (0, (-len(word)) % (T-1)))]
```

```
phash, ptrace = Poseidon().hash(hash_input)
```

```
X = Polynomial.X()
```

```
def next_power_of_two(n: int) -> int:
    if n <= 0:
        return 1
    # If n is already a power of two, return n
    if (n & (n - 1)) == 0:
        return n
    power = 1
    while power < n:
        power <<= 1
    return power

def hash_input_index(i,j):
    if (i % (r_f+r_p) == 0 and j < 7 and i//(r_f+r_p) < len(hash_input)//7):
        return hash_input[7*(i//(r_f+r_p))+j]
    else:
        return FieldElement(0)
```

```
ptrace_domain_size = next_power_of_two(len(ptrace))
gp = FieldElement.generator() ** (3*2**30/(ptrace_domain_size))
x = [gp**i for i in range(len(trace['i']))]
z = [gp**i for i in range(len(ptrace))]
```

File display

```
state_polynomial = interpolate_poly(x, [FieldElement(automata.state_map[s]) for s in trace['s']])
caracter_polynomial = interpolate_poly(x, [FieldElement(automata.caracter_map[c]) for c in trace['c']])
transi_x = [FieldElement(automata.state_map[s]*automata.caracter_map[c]) for s,c in product(automata.states, automata.alphabet)]
transition_polynomial = interpolate_poly(transi_x, [FieldElement(automata.state_map[automata.transition[s][c]]) for (s,c) in product(automata.states, a
step_polynomial = state_polynomial(gp*X) - transition_polynomial(state_polynomial(X)*caracter_polynomial(X))
ptrace_polynomials = [interpolate_poly(z, [row[i] for row in ptrace]) for i in range(0,len(ptrace[0]))]
arc_polys = [interpolate_poly(z, [FieldElement(ARC[j%(r_f+r_p)][i]) for j in range(len(ptrace))]) for i in range(len(ARC[0]))]
hash_input_poly = [interpolate_poly(z, [hash_input_index(i,j) for i in range(len(z))]) for j in range(len(ptrace[0]))]
```

Annexe: Code

```
w = FieldElement.generator()
h = FieldElement.generator() ** (3*2**30/(ptrace_domain_size*blowup_factor))
H = [h**i for i in range(ptrace_domain_size*blowup_factor)]
eval_domain = [w*h for h in H]
```

```
state_eval = [state_polynomial(d) for d in tqdm(eval_domain)]
caracter_eval = [caracter_polynomial(d) for d in tqdm(eval_domain)]
#transition_eval = [transition_polynomial(state*caracter) for state,caracter in tqdm(product(state_eval,caracter_eval))]
transition_eval = [transition_polynomial(state*caracter) for state,caracter in zip(state_eval,caracter_eval)]
ptrace_evals = [[p(x) for x in tqdm(eval_domain)] for p in ptrace_polynomials]
hash_evals = [[p(x) for x in tqdm(eval_domain)] for p in hash_input_poly]
arc_polys_eval = [[p(x) for x in tqdm(eval_domain)] for p in arc_polys]
```

Annexe: Code

```
state_merkle = MerkleTree(state_eval)
caracter_merkle = MerkleTree(caracter_eval)
transition_merkle = MerkleTree(transition_eval)
ptrace_merkle = MerkleTree([e for p in ptrace_evals for e in p])
hash_merkle = MerkleTree([e for p in hash_evals for e in p])
arc_merkle = MerkleTree([e for p in arc_polys_eval for e in p])
```

```
initial_constraint_p = (state_polynomial-automata.state_map['q0'])/(X-1)
Z_G = Polynomial([FieldElement.one()])
for i in range(len(x)):
  Z_G = Z_G * (X-x[i])
ap = Polynomial([FieldElement.one()])
for c in automata.alphabet:
    ap = ap * (caracter_polynomial - automata.caracter_map[c])
caracter_constraint_p = ap/Z_G
ap = Polynomial([FieldElement.one()])
for f in automata.accept_states:
  ap = ap * (state_polynomial - automata.state_map[f])
accepting_constraint_p = ap/(X-x[-1])
step_constraint_p = step_polynomial/(Z_G/(X-x[-1]))
poseidon_full_round_polynomials = mds@[(ptrace_polynomials[j](X) + hash_input_poly[j](X) + arc_polys[j](X))**5 for j in range(T)]
poseidon_partial_round_polynomials = mds@[(ptrace_polynomials[j](X) + hash_input_poly[j](X) + arc_polys[j](X))**(5 if j == 0 else 1) for j in range(T)]
X_G = Polynomial([FieldElement.one()])
Z_G = Polynomial([FieldElement.one()])
for i in range(len(ptrace)-1):
    r = i%(r_f+r_p)
    if(r < r_f/2 or r >= r_f/2+r_p):
      X_G = X_G * (X-z[i])
    else:
        Z_G = Z_G * (X-z[i])
poseidon_constraint_full_round = [(ptrace_polynomials[j](gp*X) - poseidon_full_round_polynomials[j](X))/X_G for j in range(len(ptrace_polynomials))]
poseidon_constraint_partial_round = [(ptrace_polynomials[j](gp*X) - poseidon_partial_round_polynomials[j](X))/Z_G for j in range(len(ptrace_polynomials
poseidon_constraints_initial = [p/(X-1) for p in ptrace_polynomials]
poseidon_output_constraint = (ptrace_polynomials[0](X)-phash[0])/(X-z[-1])
```

Annexe: Code

```
channel = Channel()
channel.send(str(ptrace_domain_size))
channel.send(str(len(ptrace)))
channel.send(str(len(eval_domain)))
channel.send(str(len(trace['i'])))
channel.send(str(phash[0]))
channel.send(state_merkle.root)
channel.send(caracter_merkle.root)
channel.send(transition_merkle.root)
channel.send(ptrace_merkle.root)
channel.send(hash_merkle.root)
channel.send(arc_merkle.root)
```

```
ps = [poseidon_output_constraint, initial_constraint_p, caracter_constraint_p, accepting_constraint_p, step_constraint_p, *poseidon_constraints_initial
alphas = [channel.receive_random_field_element() for _ in range(len(ps))]
cp = reduce(lambda x, y: x+y,map(lambda a, p: a*p, alphas, ps),FieldElement(0))
```

```
cp_eval = [cp(x) for x in eval_domain]
cp_merkle = MerkleTree(cp_eval)
channel.send(cp_merkle.root)
```

```
def fold_domain(domain):
  return [x**2 for x in domain[:len(domain)//2]]

def fold(p, beta):
  odd = p.poly[1::2]
  even = p.poly[::2]
  o = Polynomial(odd)
  e = Polynomial(even)
  return e + beta*o

def next_fri_layer(poly, domain, beta):
    next_poly = fold(poly, beta)
    next_domain = fold_domain(domain)
    next_layer = [next_poly(x) for x in next_domain]
    return next_poly, next_domain, next_layer
```

File display

```
cp.degree(), len(cp_eval)
```

Annexe: Code

```python
def FriCommit(cp, domain, cp_eval, cp_merkle, channel):
    fri_polys = [cp]
    fri_domains = [domain]
    fri_layers = [cp_eval]
    fri_merkles = [cp_merkle]
    betas = []
    i = 0
    while fri_polys[-1].degree() > 0:
        beta = channel.receive_random_field_element()
        betas.append(beta)
        next_poly, next_domain, next_layer = next_fri_layer(fri_polys[-1], fri_domains[-1], beta)
        fri_polys.append(next_poly)
        fri_domains.append(next_domain)
        fri_layers.append(next_layer)
        fri_merkles.append(MerkleTree(next_layer))
        channel.send(fri_merkles[-1].root)
        i+=1
    channel.send("FINISHED_FRI")
    channel.send(str(fri_polys[-1].poly[0]))
    return fri_polys, fri_domains, fri_layers, fri_merkles, betas
```

```python
fri_polys, fri_domains, fri_layers, fri_merkles, betas = FriCommit(cp, eval_domain,cp_eval,cp_merkle,channel)
```

```python
len(fri_layers[-1])
```

```python
def decommit_on_fri_layers(idx, channel):
    prev_idx = None
    prev_sibidx = None
    p_idx = None
    i = 0
    for layer, merkle in zip(fri_layers[:-1], fri_merkles[:-1]):
        length = len(layer)
        idx = idx % length
        sib_idx = (idx + length // 2) % length
        channel.send(str(layer[idx]))
        channel.send(str(merkle.get_authentication_path(idx)))
        channel.send(str(layer[sib_idx]))
        channel.send(str(merkle.get_authentication_path(sib_idx)))
    channel.send(str(fri_layers[-1][0]))
```

Annexe: Code

```
def decommit_on_query(idx, channel):
    assert idx + blowup_factor < len(eval_domain), f'query index: {idx} is out of range. Length of layer: {len(eval_domain)}.'
    channel.send(str(state_eval[idx%len(eval_domain)])) # f(x).
    channel.send(str(state_merkle.get_authentication_path(idx%len(eval_domain)))) # auth path for f(x).
    assert(state_eval[(idx + blowup_factor)%len(eval_domain)] == state_polynomial(gp*eval_domain[idx%len(eval_domain)]))
    channel.send(str(state_eval[(idx + blowup_factor)%len(eval_domain)])) # f(gx).
    channel.send(str(state_merkle.get_authentication_path((idx + blowup_factor)%len(eval_domain)))) # auth path for f(gx).
    assert (idx + blowup_factor)%len(eval_domain) < len(caracter_eval), f'query index: {idx%len(eval_domain)} is out of range. Length of layer: {len(ca
    channel.send(str(caracter_eval[idx%len(eval_domain)])) # f(x).
    channel.send(str(caracter_merkle.get_authentication_path(idx%len(eval_domain)))) # auth path for f(x).
    assert (idx + blowup_factor)%len(eval_domain) < len(transition_eval), f'query index: {idx%len(eval_domain)} is out of range. Length of layer: {len(
    transi_x = transition_polynomial(caracter_eval[idx%len(eval_domain)]*state_eval[idx%len(eval_domain)])
    channel.send(str(transi_x)) # f(x).
    transi_idx = (len(eval_domain)*idx+idx)%len(eval_domain)
    channel.send(str(transition_merkle.get_authentication_path(transi_idx))) # auth path for f(x).


    #### POSEIDON ####
    for row in range(len(ptrace[0])):
        trace_x = ptrace_evals[row][idx%len(eval_domain)]
        channel.send(str(trace_x))
        channel.send(str(ptrace_merkle.get_authentication_path((len(eval_domain)*row+(idx%len(eval_domain))))))
        assert(ptrace_polynomials[row](eval_domain[idx%len(eval_domain)]) == trace_x)

        next_trace_x = ptrace_evals[row][(idx+blowup_factor)%len(eval_domain)]
        channel.send(str(next_trace_x))
        channel.send(str(ptrace_merkle.get_authentication_path((len(eval_domain)*row+((idx+blowup_factor)%len(eval_domain))))))
        assert(ptrace_polynomials[row](gp*eval_domain[idx%len(eval_domain)]) == next_trace_x)

        hash_x = hash_evals[row][idx%len(eval_domain)]
        channel.send(str(hash_x))
        channel.send(str(hash_merkle.get_authentication_path((len(eval_domain)*row+(idx%len(eval_domain))))))

        arc_x = arc_polys_eval[row][idx%len(eval_domain)]
        channel.send(str(arc_x))
        channel.send(str(arc_merkle.get_authentication_path((len(eval_domain)*row+(idx%len(eval_domain))))))

    decommit_on_fri_layers(idx, channel)
```

```
def decommit_fri(channel):
    for query in range(N_QUERY):
        decommit_on_query(channel.receive_random_int(0, len(eval_domain)), channel)
```

```
decommit_fri(channel)
```

File display

Annexe: Code

```
proof_file = open("proof.txt", "w")
proof_file.write(str(channel.proof))
```

```
print_prover_success_banner(start_all, channel, len(ptrace), len(state_eval))
```

Annexe: Code

```
import ast
import constants
from constants import blowup_factor
from channel import Channel
from field import FieldElement
from merkle import verify_decommitment
from poseidon import generate_cauchy_matrix
from functools import reduce
from utils import Automata
```

Annexe: Code

```python
import time
import sys

def print_success_banner(start_time, channel, EVAL_DOMAIN_LENGTH, POSEIDON_TRACE_EVAL_LENGTH):
    # 1. Calculate Duration
    duration = time.time() - start_time

    # 2. Calculate Proof Size
    # We sum the bytes of all strings sent in the proof
    proof_size_bytes = sum(len(entry.encode('utf-8')) for entry in channel.proof)
    proof_size_kb = proof_size_bytes / 1024

    # 3. ASCII Art & Colors
    GREEN = "\033[92m"
    CYAN = "\033[96m"
    YELLOW = "\033[93m"
    RESET = "\033[0m"
    BOLD = "\033[1m"

    ascii_art = f"""
{GREEN}{BOLD}
  _____  _____   ____   ____  ______   __      __   _      _____  _____
 |  __ \|  __ \ / __ \ / __ \|  ____|  \ \    / /\ | |    |_   _||  __ \
 | |__) | |__) | |  | | |  | | |__      \ \  / /  \ | |      | |  | |  | |
 |  ___/|  _  /| |  | | |  | |  __|      \ \/ / /\ \| |      | |  | |  | |
 | |    | | \ \| |__| | |__| | |          \  / ____ \ |____ _| |_ | |__| |
 |_|    |_|  \_\\____/ \____/|_|           \/_/    \_\_____|_____|_____/

{RESET}"""

    print(ascii_art)
    print(f"{BOLD}Status:{RESET}   {GREEN}VERIFICATION SUCCESSFUL{RESET}")
    print("-" * 50)
    print(f"{BOLD}Metrics:{RESET}")
    print(f"  ⏱️  {CYAN}Time Taken:{RESET}    {duration:.4f} seconds")
    print(f"  📦 {YELLOW}Proof Size:{RESET}    {proof_size_kb:.2f} KB")
    print(f"  🔐 {CYAN}Queries:{RESET}       {constants.N_QUERY}")
    max_constraint_degree = (POSEIDON_TRACE_EVAL_LENGTH - 1) * 5

    # 2. Calculate Probability of Cheating (Soundness Error)
    # Rho = Degree / Domain_Size
    rho = max_constraint_degree / EVAL_DOMAIN_LENGTH
    p_cheat = rho ** constants.N_QUERY

    # 3. Print Code
    print(f"  🎲 {CYAN}P(Cheating):{RESET}    {p_cheat:.8f} (approx. 1 in {int(1/p_cheat)})")
    print("-" * 50)

    # Optional: Contextualize for the Jury
    print(f"{BOLD}Interpretation:{RESET}")
    if proof_size_kb < 200:
        print("  ✅ Succinct Proof (Very small size!)")
    else:
        print("  ⚠️ Large Proof (Consider checking trace/blowup)")
```

Annexe: Code

```
proof_file = open("proof.txt", "r")
proof = ast.literal_eval(proof_file.read())
automata = Automata.from_json(open("automata.json", "r").read())
```

```
def parse_proof(method, parse = None):
    global v_current_channel_idx
    global v_proof
    #print("channel",proof[current_channel_idx], "method", method)
    v_current_channel_idx+=1
    if parse == None:
        return v_proof[v_current_channel_idx-1][len(method)+1:]
    else:
        return parse(v_proof[v_current_channel_idx-1][len(method)+1:])
```

Annexe: Code

```
channel = Channel(proof)
v_proof = channel.proof
v_c = Channel()
v_current_channel_idx = 0
PTRACE_DOMAIN_SIZE=parse_proof("send", lambda x: int(x))
POSEIDON_TRACE_EVAL_LENGTH=parse_proof("send", lambda x: int(x))
EVAL_DOMAIN_LENGTH=parse_proof("send", lambda x: int(x))
AUTOMATA_TRACE_LENGTH=parse_proof("send", lambda x: int(x))
v_hash = v_proof[v_current_channel_idx][5:]
v_c.send(v_hash)
v_current_channel_idx+=1
v_state_merkle = v_proof[v_current_channel_idx][5:]
v_c.send(v_state_merkle)
v_current_channel_idx+=1
v_caracter_merkle = v_proof[v_current_channel_idx][5:]
v_c.send(v_caracter_merkle)
v_current_channel_idx+=1
v_transition_merkle = v_proof[v_current_channel_idx][5:]
v_c.send(v_transition_merkle)
v_current_channel_idx+=1
v_ptrace_merkle = v_proof[v_current_channel_idx][5:]
v_c.send(v_ptrace_merkle)
v_current_channel_idx+=1
v_hash_merkle = v_proof[v_current_channel_idx][5:]
v_c.send(v_hash_merkle)
v_current_channel_idx+=1
v_arc_merkle = v_proof[v_current_channel_idx][5:]
v_c.send(v_arc_merkle)
v_current_channel_idx+=1
v_alphas = []
for i in range(constants.CONSTRAINTS_LENGTH):
    v_a = FieldElement(int(v_proof[v_current_channel_idx][len('receive_random_field_element:'):]))
    v_current_channel_idx+=1
    v_alphas.append(v_a)
#a0 = FieldElement(int(proof[current_channel_idx][len('receive_random_field_element:'):]))
#assert(c.receive_random_field_element() == a0)
#current_channel_idx+=1
#a1 = FieldElement(int(proof[current_channel_idx][len('receive_random_field_element:'):]))
#assert(c.receive_random_field_element() == a1)
#current_channel_idx+=1
#a2 = FieldElement(int(proof[current_channel_idx][len('receive_random_field_element:'):]))
#assert(c.receive_random_field_element() == a2)
#current_channel_idx+=1
#a3 = FieldElement(int(proof[current_channel_idx][len('receive_random_field_element:'):]))
#assert(c.receive_random_field_element() == a3)
#current_channel_idx+=1
#a4 = FieldElement(int(proof[current_channel_idx][len('receive_random_field_element:'):]))
#assert(c.receive_random_field_element() == a4)
#current_channel_idx+=1
cp_merkle_root = v_proof[v_current_channel_idx][len('send:'):]
v_current_channel_idx+=1
```

File display

Annexe: Code

```
w = FieldElement.generator()
h = FieldElement.generator() ** (3*2**30/(PTRACE_DOMAIN_SIZE*blowup_factor))
gp = FieldElement.generator() ** (3*2**30/(PTRACE_DOMAIN_SIZE))

#def x(i):
#    return gp**i

def x(i):
    return gp**(i%AUTOMATA_TRACE_LENGTH)

def z(i):
    return gp**(i%POSEIDON_TRACE_EVAL_LENGTH)

#x = [gp**i for i in range(AUTOMATA_TRACE_LENGTH)]
#z = [gp**i for i in range(POSEIDON_TRACE_EVAL_LENGTH)]
mds = generate_cauchy_matrix(constants.T)
```

```
def verify_FRI_commitment():
    global v_proof
    global v_current_channel_idx
    v_betas = []
    v_roots = [cp_merkle_root]
    v_l = 0
    while v_proof[v_current_channel_idx] != "send:FINISHED_FRI":
        v_betas.append(FieldElement(int(v_proof[v_current_channel_idx][len('receive_random_field_element:'):])))
        v_current_channel_idx+=1
        v_roots.append(v_proof[v_current_channel_idx][len('send:'):])
        v_current_channel_idx+=1
        v_l+=1
    v_current_channel_idx+=1
    v_poly0 = int(v_proof[v_current_channel_idx][len('send:'):])
    v_current_channel_idx+=1
    return v_betas,v_roots,v_poly0,v_l
```

```
v_betas,v_roots,v_poly0,v_l = verify_FRI_commitment()
```

```
print(v_l), 2**10
```

Annexe: Code

```python
def verify_query_decommitment(l):
    global current_channel_idx
    for query in range(constants.N_QUERY):
        # r = int(proof[current_channel_idx][len('receive_random_field_element:'):])
        r = parse_proof("receive_random_int", int)
        # current_channel_idx+=1
        state_x = parse_proof("send", lambda x: FieldElement(int(x)))
        #state_x = FieldElement(int(proof[current_channel_idx][len('send:'):]))
        #current_channel_idx+=1
        state_x_path = parse_proof("send", lambda x: ast.literal_eval(x))
        state_x2 = parse_proof("send", lambda x: FieldElement(int(x)))
        #state_x = FieldElement(int(proof[current_channel_idx][len('send:'):]))
        #current_channel_idx+=1
        state_x2_path = parse_proof("send", lambda x: ast.literal_eval(x))
        #state_x_path = ast.literal_eval(proof[current_channel_idx][len('send:'):])
        #current_channel_idx+=1
        assert(verify_decommitment(r,state_x,state_x_path,v_state_merkle))
        assert(verify_decommitment((r+blowup_factor),state_x2,state_x2_path,v_state_merkle))

        #caracter_x = FieldElement(int(proof[current_channel_idx][len('send:'):]))
        #current_channel_idx+=1
        caracter_x = parse_proof("send", lambda x: FieldElement(int(x)))
        caracter_x_path = parse_proof("send", lambda x: ast.literal_eval(x))
        #caracter_x_path = proof[current_channel_idx][len('send:'):]
        #current_channel_idx+=1
        assert(verify_decommitment(r,caracter_x,caracter_x_path,v_caracter_merkle))

        transition_x = parse_proof("send", lambda x: FieldElement(int(x)))
        transition_x_path = parse_proof("send", lambda x: ast.literal_eval(x))
        assert(verify_decommitment((EVAL_DOMAIN_LENGTH*r+r),transition_x,transition_x_path,v_transition_merkle))

        poseidon_x = []
        poseidon_path = []
        poseidon_next_x = []
        poseidon_next_path = []
        hash_x = []
        hash_path = []
        arc_x = []
        arc_path = []

        for i in range(constants.T):
            poseidon_x.append(parse_proof("send", lambda x: FieldElement(int(x))))
            poseidon_path.append(parse_proof("send", lambda x: ast.literal_eval(x)))
            assert(verify_decommitment((i*EVAL_DOMAIN_LENGTH + (r)),poseidon_x[-1],poseidon_path[-1],v_ptrace_merkle))

            poseidon_next_x.append(parse_proof("send", lambda x: FieldElement(int(x))))
            poseidon_next_path.append(parse_proof("send", lambda x: ast.literal_eval(x)))
            assert(verify_decommitment((i*EVAL_DOMAIN_LENGTH  + ((r + blowup_factor))),poseidon_next_x[-1],poseidon_next_path[-1],v_ptrace_merkle))

            hash_x.append(parse_proof("send", lambda x: FieldElement(int(x))))
            hash_path.append(parse_proof("send", lambda x: ast.literal_eval(x)))
            assert(verify_decommitment((i*EVAL_DOMAIN_LENGTH  + (r)),hash_x[-1],hash_path[-1],v_hash_merkle))

            arc_x.append(parse_proof("send", lambda x: FieldElement(int(x))))
            arc_path.append(parse_proof("send", lambda x: ast.literal_eval(x)))
            assert(verify_decommitment((i*EVAL_DOMAIN_LENGTH  + (r)),arc_x[-1],arc_path[-1],v_arc_merkle))
```

File display

Annexe: Code

```
#transition_x = FieldElement(int(proof[current_channel_idx][len('send:'):]))
#current_channel_idx+=1
#transition_x_path = proof[current_channel_idx][len('send:'):]
#current_channel_idx+=1

layers = []
layer_paths = []

#length = 2**(v_l+1)
length = EVAL_DOMAIN_LENGTH
cps = []

for j in range(0,l):
    idx = r % length
    sib_idx = (r + length // 2) % length
    layer_x = parse_proof("send", lambda x: FieldElement(int(x)))
    #layer_x = int(proof[current_channel_idx][len('send:'):])
    #current_channel_idx+=1
    layer_x_path = parse_proof("send",  lambda x: ast.literal_eval(x))
    assert(verify_decommitment(idx,layer_x,layer_x_path,v_roots[j]))
    #layer_x_path = proof[current_channel_idx][len('send:'):]
    #current_channel_idx+=1
    layer_sibx = parse_proof("send", lambda x: FieldElement(int(x)))
    #layer_sibx = int(proof[current_channel_idx][len('send:'):])
    #current_channel_idx+=1
    #layer_sibx_path = proof[current_channel_idx][len('send:'):]
    layer_sibx_path = parse_proof("send",  lambda x: ast.literal_eval(x))
    assert(verify_decommitment(sib_idx,layer_sibx,layer_sibx_path,v_roots[j]))
    e = (w*h**idx)**(2**j)

    g_x = (layer_x + layer_sibx)/FieldElement(2)
    h_x = (layer_x - layer_sibx)/(2*e)
    cp_ip1 = g_x + v_betas[j]*h_x
    cps.append(cp_ip1)

    if j == 0:
        initial_constraint = (state_x-automata.state_map['q0'])/(e-1)
        Z_G = FieldElement(1)
        for i in range(AUTOMATA_TRACE_LENGTH):
          Z_G = Z_G * (e-x(i))
        ap = FieldElement(1)
        for c in automata.alphabet:
            ap = ap * (caracter_x - automata.caracter_map[c])
        caracter_constraint = ap/Z_G
        ap = FieldElement(1)
        for f in automata.accept_states:
          ap = ap * (state_x - automata.state_map[f])
        accepting_constraint = ap/(e-x(-1))
        step_constraint = (state_x2 - transition_x)/(Z_G/(e-x(-1)))
        X_G = FieldElement(1)
        Y_G = FieldElement(1)
```

Annexe: Code

```
            for i in range(POSEIDON_TRACE_EVAL_LENGTH-1):
                round_x = i%(constants.r_f+constants.r_p)
                if(round_x < constants.r_f/2 or round_x >= constants.r_f/2+constants.r_p):
                  X_G = X_G * (e-z(i))
                else:
                  Y_G = Y_G * (e-z(i))
            v_poseidon_full_round = mds@[(poseidon_x[j] + hash_x[j] + arc_x[j])**5 for j in range(constants.T)]
            v_poseidon_partial_round = mds@[(poseidon_x[j] + hash_x[j] + arc_x[j])**(5 if j == 0 else 1) for j in range(constants.T)]
            poseidon_full_round_cr = [(poseidon_next_x[j] - v_poseidon_full_round[j])/X_G for j in range(constants.T)]
            poseidon_partial_round_cr = [(poseidon_next_x[j] - v_poseidon_partial_round[j])/Y_G for j in range(constants.T)]
            poseidon_initial_cr = [poseidon_x[j]/(e-1) for j in range(constants.T)]
            poseidon_output_cr = (poseidon_x[0]-FieldElement(int(v_hash)))/(e-z(-1))

            v_ps = [poseidon_output_cr, initial_constraint, caracter_constraint, accepting_constraint, step_constraint, *poseidon_initial_cr, *pose
            cp_e = reduce(lambda x, y: x+y,map(lambda a, p: a*p, v_alphas, v_ps),FieldElement(0))
            assert(cp_e == layer_x)
        else:
            assert(cps[j-1] == layer_x)

        layers.append((layer_x,layer_sibx))
        layer_paths.append((layer_x_path, layer_sibx_path))
        length //= 2

    last_layer_x = parse_proof("send", int)
    assert(cps[-1] == last_layer_x)
```

```
verify_query_decommitment(v_l)
```

```
print_success_banner(verification_start, channel, EVAL_DOMAIN_LENGTH, POSEIDON_TRACE_EVAL_LENGTH)
```

```
 ____   ____   ___   ___   _____   _      __   _     ____  ____
|  _ \|  _ \ / _ \ / _ \| ____|  \ \    / /\ | |   |_   _||  _ \
| |_) | |_) | | | | | | |  _|     \ \  / /  \ | |     | | | | | |
|  __/|  _ /| |_| | |_| | |__     \ \/ / /\ \| |___  | | | |_| |
| |   | | \ \|  _| |  _| |       \  / ___ \ |___ _| |_| |  _| |
|_|   |_|  \_\___/ \___/|_|        \/_/   \_\_____|____|____/
```

```
Status:   VERIFICATION SUCCESSFUL
--------------------------------------------------
Metrics:
  ⏱️  Time Taken:    0.2546 seconds
  📦 Proof Size:    771.66 KB
  🔐 Queries:       12
  🎲 P(Cheating):    0.00000000 (approx. 1 in 18514789709398876160)
--------------------------------------------------
Interpretation:
  ⚠️ Large Proof (Consider checking trace/blowup)
```